



Fig. 3: Pairing with Bluetooth module

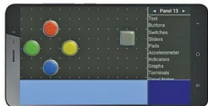


Fig. 4: Configuring the buttons

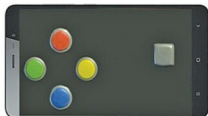


Fig. 5: Final setup in smartphone

Arduino Mega is used because it has 16 analogue channels and can be used to connect different sensors. The flow-chart (Fig. 6) depicts the algorithm used in the navigation program (`arduino_bluetooth.ino`) burnt into the MCU of Arduino.

Capturing live video. Servo motors are connected to GPIO pins of RPi. In order to control the position of the camera, two servo motors (M1 and M2) are used to move the camera in X and Y directions.

Download the Wireless IMU app from Play Store and install in your smartphone. The app supports accelerometer, gyroscope and magnetometer, and can be made to run in the background. The values from this app are sent to the RPi through UDP protocol. The magnetometer values make the servos move in either right or left direction. The accelerometer values make the servos move up and down. Only Z direction is used in the project. The algorithm for servo control is shown in Fig. 7. Thus, the camera can capture live video at different angles.

Arduino Mega 2560. Arduino Mega 2560 is a microcontroller board

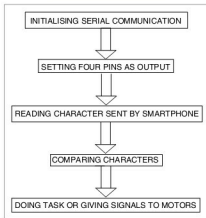


Fig. 6: Flow-chart of navigation program

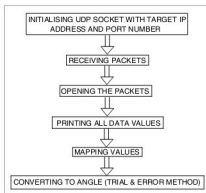


Fig. 7: Algorithm for servo motor interfacing

based on the ATmega2560. It has 54 digital input/output pins (of which 15 can be used as PWM outputs), 16 analogue inputs, four UARTs (hardware serial ports), a 16MHz crystal oscillator, a USB connection, a power jack, an ICSP header and a reset button. It contains everything needed to support the microcontroller; simply connect it to a computer with a USB cable or power it with an AC-to-DC adaptor or battery to get started. The Mega 2560 board is compatible with most shields designed for the Uno. It is an update to the Arduino Mega.

The Arduino board acts as the directional processor of the robot. Commands to drive the robot in the specified direction are processed and given by the Arduino to a driver IC, which, in turn, causes the motors to run. The Mega 2560 board can be programmed with the Arduino software (IDE). ATmega2560 on the board comes preprogrammed with a bootloader that allows you to upload new code to it without using an exter-

nal hardware programmer.

The Mega 2560 board can be powered via USB connection or with an external power supply. The power source is selected automatically. External (non-USB) power can come either from an AC-to-DC adaptor (wall-wart) or battery. The board can operate on an external supply of 6-20 volts. The recommended power supply range is 7-12 volts.

ATmega2560 has 256kB of flash memory for storing code (of which 8kB is used for the bootloader), 8kB SRAM and 4kB EEPROM. The Mega 2560 board has a number of facilities to communicate with a computer, another board or other microcontrollers. An ATmega16U2 on the board channels one of these over USB and provides a virtual COM port to software on the computer.

Gear motor. Geared motors are a specific type of electrical motors that produce a high torque while maintaining a low-horsepower or low-speed motor output. These can be either AC (alternating current) or DC (direct current). They also have two different speed specifications—normal speed and stall-speed torque.

DC geared motors are primarily used to reduce speed in a series of gears, which, in turn, creates more torque. This is accomplished by an integrated series of gears or a gear box attached to the main motor rotor and shaft via a second reduction shaft. The second shaft is then connected to the series of gears or gearbox to create what is known as a series of reduction gears.

Two DC geared motors are used in this project.

Servo motor. Servo motor is a rotatory actuator or linear actuator that allows for precise control of angular or linear position, velocity and acceleration. It consists of a suitable motor coupled to a sensor for position feedback. It also requires a relatively sophisticated controller, often a dedicated module designed specifically for use with the servo motors. It is not a specific class of motor, although the term servomotor is often used to refer to a motor suitable for use in a closed loop control system. Two servo motors are used in this project.